

COMP3018: Report (Literature Review & Programming Project)

Cognitive Robotics

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1- Task (3) Literature Review

☐ cite this lecture and book: /Users/alfienurse/Desktop/gitdev/Uni/Undergrad/year-3/Work/1st Year/Semester 2/3018-cw/learning/[] final: cognitive architecture

☒ [X] ## 1.1. Introduction

Ageing populations and a shrinking care workforce positioned **assistive robotics** (*human-supportive robots within physical, cognitive, and social realms of impairment affecting daily-living activities*); a prominent technological response to a widening care gap. The field spans physical prosthetics; surgical assistance; neurodivergent support; exoskeletal rehabilitation, and social companionship; this essay however focuses on

socially assistive robotics (*SAR*), an assistive-robot subfield wherein the robot “focuses on helping human users through social rather than physical interaction” (Tapus, Matarić and Scassellati, 2007, Abstract), as here most active research and ethical tension converge.

Robots now administer medication reminders, facilitate rehabilitation exercises, and therapeutically companionate in clinical and domestic settings: reduced caregiver burden, improved patient outcomes wherein residents became “more active and more communicative, both with each other and their caregivers” (Wada and Shibata, 2007, p. 973), and increased “social interaction” among elderly residents (Wada and Shibata, 2007, p. 972, Abstract).

However, most-current assistive robots operate at what Sciutti et al. (2023, p. 160) call the social layer: they react to immediate stimuli but lack the cognitive depth to anticipate user needs, remember past interactions, or reason about their own performance. Sciutti et al. argue that effective assistive robots should be *cognitive*: capable of “flexible, context-sensitive action, knowing what they are doing and why they are doing it.” Vernon, Metta and Sandini (2007, p. 151) formalise this requirement via a “virtuous cycle that is embedded in an ongoing process of action and perception” (the agent anticipates → learns → adapts to achieve autonomy). This essay contends that assistive robotics should graduate from reactive social behaviour to cognitive capability (intelligence deployed *over* the social layer) if it is to deliver sustained, personalised support. The following sections survey the theoretical foundations thereof: evaluate prominent applications through this cognitive lens, discuss challenges and ethical implications, and identify future directions.

[X] ## 1.2. Literature Review

Cognitive robotics: defined by Sciutti et al. (2023, p. 160), lies at the intersection of Robotics, Artificial Intelligence, and Cognitive and Biological Sciences, combining “sensorimotor behaviour, higher-level functions, and social capabilities of an intelligent robot.” This interdisciplinary grounding distinguishes it from conventional robotics (*treats the robot as purely engineered*) and from social robotics (*addresses interaction behaviour without necessarily modelling cognitive processes*). The distinction is consequential: a robot that smiles when a patient smiles is social; a robot that infers *why* the patient is smiling, and adjusts its future strategy accordingly, is therefore *cognitive*.

□ verify following lecture slides

Vernon, Metta and Sandini (2007, p. TODO) synthesise the field’s definitional plurality into a core cycle. The European Network for Advancement of Artificial Cognitive Systems (euCognition) catalogued 42 definitions of cognition, yet the common thread therein is: anticipation, learning, and adaptation, intersected with perception and action to create autonomy (Fig. 1). This cycle provides an architectural checklist for assistive robots: a system that cannot direct its gaze toward relevant stimuli whilst suppressing irrelevant ones (*selective attention*), anticipate the outcome of its actions (*i.e. prospection*), learn from past interactions (*memory*), or adapt its strategy when performance declines (*metacognition*) is, per this framework, not yet cognitive. Sciutti et al. (2023, TODO: verify p. 160) further specify that cognitive robots should “reason about their actions and modify their behavior to improve their effectiveness”; a capacity termed *theory of mind*, wherein the agent infers another’s hidden mental state from observable behaviour.

Furthermore, memory is not monolithic. Vernon, Metta and Sandini (2007, p. TODO) distinguish *episodic memory* (*records of specific past experiences and their contextual outcomes*) from *semantic memory* (*general knowledge about the world, including spatial relationships and factual constraints*); a typology Laird (2012, p. 225) crystallises in implementational terms: episodic memory is “what you ‘remember’” whilst semantic memory is “what you ‘know’”. For example, assistive-medication robots need episodic memory to recall that a user refused medication after a restless night, and semantic memory to know certain drugs cannot also be administered. Whilst the 42-definitions problem confirms the field lacks consensus on what cognition *per se* is, the common thread (anticipation, learning, adaptation) is exactly what assistive robotics demands.

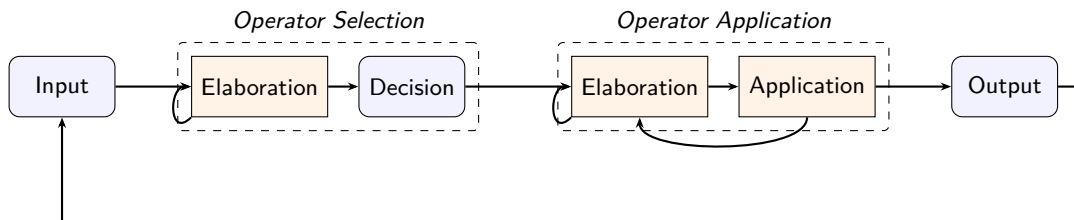


Figure 1: Soar’s processing cycle, adapted from Laird (2012, fig. 4.7, p. 79), as a concrete instantiation of Vernon, Metta and Sandini’s (2007) cognition cycle. Laird specifies that the cycle “consists of four phases: Input, Operator Selection, Operator Application, and Output” (Laird, 2012, p. 79), wherein Input maps to perception, Operator Selection draws on episodic and semantic memory to anticipate consequences (i.e. *prospection*), Operator Application executes the chosen behaviour through parallel rule-firing waves, and Output returns the agent to the environment whilst the loop hands control back to Input for continual adaptation. This architecture exposes the deficit of reactive systems: PARO, for instance, implements only Input $\rightarrow$ Output, lacking the substate-driven Operator Selection wherein deliberation over latent user states could occur. Furthermore, Laird argues that “the most significant effect of chunking is that it eliminates processing in substates for situations similar to ones experienced in the past” (Laird, 2012, p. 164), wherein repeated deliberations compile into procedural rules; a mechanism distinct from POMDP belief-space approximation, yet complementary thereto in addressing real-time embodied operation under care-time constraints.

□ ## 1.3. Applications

[] 1.3.1 Therapeutic and Emotional Support

The PARO therapeutic seal robot represents one of the most-widely deployed platforms within socially assistive robotics (Tapus, Matarić and Scassellati, 2007, p. TODO). Wada and Shibata (2007, p. 974) demonstrate that PARO reduces agitation and improves mood in patients with dementia, utilising tactile sensors and auditory processing to modulate its behaviour in response to touch and voice. Clinical trials report that urinary stress indicators “significantly improved” after PARO’s introduction (Wada and Shibata, 2007, TODO: verify: p. 978, Table II), therefore the platform has been adopted in care homes across Japan, Europe, and the United States.

Notwithstanding the benefits, PARO operates at the reactive layer: it possesses no theory of mind (it cannot infer *why* a patient is agitated (loneliness, pain, confusion) nor episodic memory of *what* calmed the patient previously. A cognitively-equipped therapeutic robot, in contrast, would anticipate mood shifts via *prospection* $\rightarrow$ recall that music soothed this patient yesterday via episodic memory $\rightarrow$ adapt its strategy via metacognition. Insofar as PARO’s effectiveness plateaus because it cannot personalise its responses over time, the cognitive gap is not just theoretical but potentially clinically consequential. The architectural cost is precise: without episodic memory, the agent’s perception is “limited both spatially and temporally—that is, to the here and now” (Laird, 2012, p. 233), wherein each session resets to the immediate present. Fong, Nourbakhsh and Dautenhahn (2003, p. TODO-verify 145) formalise this gap via Breazeal’s taxonomy: PARO occupies the “social interface” level (human-like cues but “shallow models of social cognition”), whereas Sciutti et al.’s (2023, TODO verify: p. 160) vision of robots “knowing what they are doing and why” demands the *socially intelligent* level. The distance between these levels is the cognitive deficit assistive robotics should close.

[] 1.3.2 Medication Adherence and Daily Living Support

Medication non-adherence imposes substantial costs on healthcare systems, and elderly patients with polypharmacy regimens are particularly vulnerable to missed or incorrect doses. Robots in this domain face a different challenge from therapeutic companionship: trust and cognitive load are latent variables that cannot be directly measured, only inferred from noisy behavioural proxies. Lee and See (2004, p. 54) define trust as “the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability”; a definition foregrounding the unobservable nature that necessitates

probabilistic modelling. A user may comply with a medication prompt despite low trust (e.g. time pressure), or indeed refuse despite high trust (e.g. task complexity), and thus the observation alone cannot reliably disambiguate the underlying state (Hancock et al., 2011, p. 522).

The Partially Observable Markov Decision Process (POMDP) provides formal machinery for this uncertainty. Chen et al. (2020, p. 6) demonstrate a Trust-POMDP wherein the robot maintains a belief distribution over trust and selects actions that maximise long-term collaboration, showing belief-space planning outperforms fixed strategies in the tested collaborative scenario. Garcez and Lamb (2023, p. 12389) identify the neuro-symbolic paradigm as the ‘third wave’ of AI, wherein neural subsystems (e.g. large language models) handle perception whilst symbolic subsystems (e.g. POMDPs) govern temporal reasoning, providing the temporal scaffold stateless systems lack. Nikolaidis, Hsu and Srinivasa (2017, p. 625) provide empirical corroboration: in a collaborative task ($n = 69$), robots utilising mutual adaptation via a Mixed Observability MDP (modelling human adaptability as a latent variable) were rated significantly more trustworthy than fixed-policy alternatives ($U = 180$, $p = 0.048$). This aligns with Hancock et al.’s (2011, p. 522) finding that robot performance attributes are the strongest trust predictors, whilst demonstrating that belief-space planning as advocated by Chen et al. (2020, p. 6) translates into measurable trust gains.

[] 1.3.3 Physical Rehabilitation and Mobility

Robotic exoskeletons and assistive manipulators for stroke recovery and mobility support constitute a third application domain. These systems need to adapt in real time not only to the patient’s physical state (joint angles and muscle activation patterns) but also to their psychological state: motivation, frustration, and fatigue are internal variables that determine whether a patient perseveres or disengages.

Embodied cognition becomes essential. Brooks (1991, p. TODO) argues that intelligence emerges from physical interaction with the environment rather than abstract representation, whereas Fong, Nourbakhsh and Dautenhahn (2003, p. 149) operationalise this as “perturbatory coupling”: the more channels of mutual influence between robot and environment, the more embodied the system. A rehabilitation robot therefore occupies a uniquely cognitive niche, as it should sense the patient’s body, reason about current capabilities, and adapt appropriately. A purely language-based or screen-based interface cannot achieve this; Matarić et al. (2007, p. TODO) confirm as much empirically, finding that stroke survivors engaged more enthusiastically with a physically embodied assistive robot than with screen-based alternatives. Tapus, Țăpuș and Matarić (2008, p. TODO), in fact show that embodiment alone is insufficient: adaptive personality matching (adjusting interaction distance and speed to the user’s traits) further improved task performance, suggesting rehabilitation robots require not just physical presence but cognitive adaptation to the individual. The cognitive building blocks required (haptic perception, prospective planning of difficulty, episodic memory of the patient’s trajectory) thus suggest an embodied cognitive architecture is necessary rather than a disembodied controller.

□ ## 1.4. Discussion

[] 1.4.1 Challenges

Tapus, Matarić and Scassellati (2007, .pdf-p. 6) projected that by 2012 SAR systems would demonstrate “marked improvement in learning/training/recovery of the user”; yet PARO, the most-deployed platform nearly twenty years later, *still* cannot remember yesterday’s session. Three challenges explain this stalled trajectory. Firstly, computational intractability: solving *POMDPs* exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987, p. 448 {TODO VERIFY}), and the belief simplex grows exponentially with state-space dimensionality. Whilst approximate solvers such as point-based value iteration (Pineau, Gordon and Thrun, 2003, p. 1025; Kaelbling, Littman and Cassandra, 1998, p. 120) and online Monte-Carlo tree search (Silver and Veness, 2010, p. 1) mitigate this, real-time cognitive processing within embodied systems remains an open challenge, particularly when multiple unobserved variables (trust, load, emotion) require tracking simultaneously.

Secondly, the measurement problem: trust, cognitive load, and emotional state are not directly observable; observations thereof are noisy proxies at best. Hancock et al.’s (2011, p. 522) meta-analysis of 29 studies finds that even the strongest correlates of trust explain only modest variance, whilst Broadbent, Stafford

and MacDonald (2009, p. TODO) note that acceptance itself depends on matching robot behaviour to user expectations rather than trust alone. Desai et al. (2013, p. 256) further demonstrate that trust dynamics are non-linear, building slowly through consistent performance but degrading rapidly after errors; and thus a single misclassified observation can cascade into inappropriate action selection. Nikolaidis, Hsu and Srinivasa (2017, p. 627), however, demonstrate that mutual adaptation partially mitigates this fragility: when the robot models human adaptability as a latent variable, trust persists even during strategy disagreements, suggesting the variance Hancock et al. report may stem from studies that treat the human as a static rather than co-adaptive partner.

Finally, adaptation without exploitation: a robot that runs inference on cognitive load could, in principle, time its medication requests to coincide with periods of high vulnerability, thereby maximising compliance at the expense of user autonomy. The reward function governing the POMDP’s policy should therefore encode ethical constraints alongside clinicians’ objectives.

[] 1.4.2 Ethical Implications

Assistive robots operating in intimate care spaces (bedrooms, bathrooms, rehabilitation clinics) continuously collect sensitive behavioural data. Facial expressions, vocal patterns, and movement trajectories constitute biometric data, yet regulatory frameworks have not kept pace with deployment. Wachter, Mittelstadt and Floridi (2017, p. TODO) argue that even the General Data Protection Regulation provides no enforceable “right to explanation” of automated decisions; a gap particularly concerning in healthcare wherein recommendations directly affect patient wellbeing.

Moreover, over-reliance on assistive robots risks eroding functional independence. If a robot consistently anticipates and pre-empts needs via prospection, the user may disengage from self-directed activity, thereby creating a dependency that contradicts the assistive mandate. Sharkey (2014, p. 6 (VERIFY)) frames this via Nordenfelt’s ‘Dignity of Identity’: “a robot that dealt impersonally with an older person, without knowing or using their name or their preferences would also be likely to negatively affect their feelings of dignity.” This implies that only cognitively-equipped robots (those with episodic memory of individual users) can avoid dignity violations; reactive systems such as PARO, regardless of their therapeutic benefits (Wada and Shibata, 2007, p. 974), risk infantilisation precisely because they cannot personalise. The responsibility gap compounds this further: when a care robot administers incorrect medication, liability falls ambiguously between manufacturer, deployer, and clinician.

Furthermore, the deployment of assistive robots is not equitable: wealthy nations with sufficient infrastructure and research investment stand to benefit, whilst low-income populations face a widening digital divide in access to care technologies. The question of whether assistive robots displace human carers or augment them remains unresolved.

The ethical watchword is therefore proactive regulation: design-stage ethics that anticipate failure modes before deployment, rather than reactive patchwork after harm. Per the embodied cognition thesis, if intelligence indeed requires a body, and that body enters the most intimate spaces of vulnerable persons, then the ethical stakes of assistive cognitive robotics are uniquely high.

□ ## 1.5. Conclusion

Assistive robotics stands at an inflection point. Current systems (PARO, medication prompt robots, rehabilitation aids) deliver measurable benefits within narrow operational envelopes, yet their reactive architectures limit sustained, personalised effectiveness. The Vernon, Metta and Sandini (2007) cognition cycle provides the architectural blueprint for graduating beyond this plateau: assistive robots that anticipate (prospection), remember (episodic and semantic memory), reason about others’ mental states (theory of mind), and monitor their own performance (metacognition) would constitute a qualitative advance over the most-capable systems deployed.

The neuro-symbolic paradigm offers a viable path toward this vision, as the Trust-POMDP framework attests (Chen et al., 2020). Sciutti et al. (2023, pp. 162-163) independently identify the integration of learning with model-based approaches as cognitive robotics’ most-prominent trajectory; that this converges with Garcez and Lamb’s (2023, p. 12389) ‘third wave’ thesis from AI theory suggests the direction is robust

rather than parochial. Future applications will likely extend beyond single-task assistance toward cognitively autonomous home-dwelling companions: robots that proactively monitor health indicators, anticipate daily needs via episodic memory, and adapt their interaction style to the user’s evolving cognitive and emotional state. Sharkey and Sharkey (2012, p. 27) identify this trajectory whilst cautioning that such systems risk replacing rather than supplementing human-care, and therefore the field should pursue cognitive capability and ethical governance in concert, lest the technology displace these very carers it was meant to supplement. Figure~2 visualises this trajectory. The ultimate test, per the embodied cognition thesis, is a robot that can: sense → remember → anticipate → adapt within the physical world, whilst respecting the autonomy, and dignity, of the persons it serves.

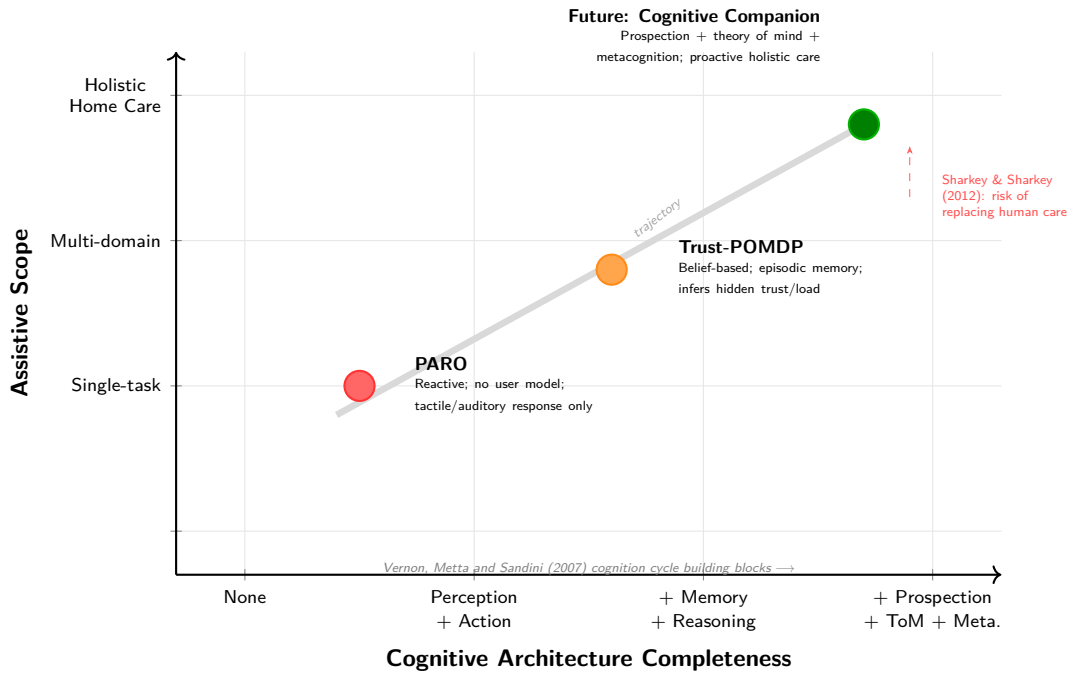


Figure 2: Trajectory of assistive robotics from reactive single-task systems (PARO) through adaptive belief-based architectures (Trust-POMDP) toward cognitively autonomous home-dwelling companions. Expanding assistive scope without expanding cognitive capability is insufficient; the diagonal trajectory requires both.

1.6. Task-3 References

1.6.1 Papers and Journal Articles

- ☐ verify all links work and extract each papers insights

REMOVE: one checkmark for working link second for insight gathered

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2- Task (4) Novel Programming Project

- ☐ CRITICAL: reconfigure report below to match newest gaze.py
- ☒ multi-layer defence-in-depth: talk about how silero-vad was used to stop whisper hallucinations
- ☐ verify word count

2.1. Introduction (10%; Salman)

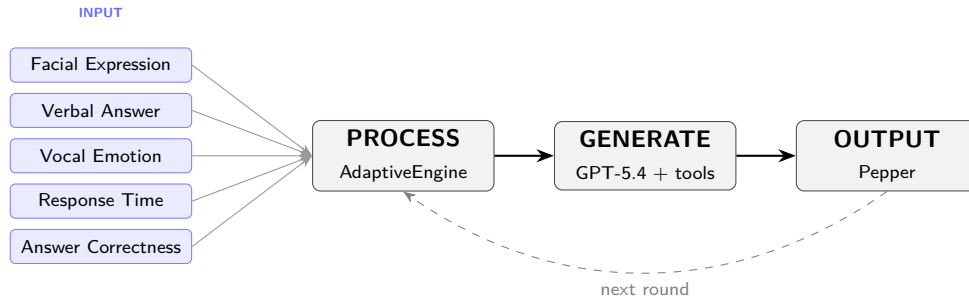


Figure 3: GAZE system architecture. Five live input signals feed the AdaptiveEngine (the symbolic reasoning layer), wherein a context block is built and passed to GPT-5.4 function-calling; a separate deterministic GPT-4.1 call handles answer verification (i.e. the referee, not the host). The generated response delivers concurrently via Pepper’s speech, gesture, and LED subsystems, whilst the dashboard pulls frames from Pepper’s ALVideoDevice socket stream rather than still photos. The loop circulates to the next round, now adapted.

As noted in {video of easy questions} wherein Salman purposely disengages from the human-robot interaction, and GAZE-Pepper

2.2. Background (10%; Alfie)

GAZE sits within socially assistive robotics (*the deployment of robots to support users through social interaction rather than physical contact*) and affective computing, which Picard (1997, p. 1, Abstract) founds as “computing that relates to, arises from, or influences emotions”; Tapus, Matarić and Scassellati (2007, p. 35) define this sub-field as systems that “assist users through social interaction.” Most-current platforms react to a single input signal (Fong, Nourbakhsh and Dautenhahn, 2003, p. 148), suffering the single-signal problem: a facial-expression classifier misreads resting faces as displeasure; a response-time metric mistakes deliberation for disengagement. Calvo and D’Mello (2010, p. 28) identify “the inherent challenges with unisensory affect detection”; Poria et al. (2017, p. 99) report that multimodal systems were “consistently (85% of systems) more accurate than their best unimodal counterparts, with an average improvement of 9.83%.” Spezialetti, Placidi and Rossi (2020, pp. 1-2, Introduction) corroborate this within HRI specifically,

reviewing recognition systems across facial, vocal, and physiological channels. This demands multi-signal fusion (the system weighs complementary channels together instead of trusting any one alone).

- PROOFREAD GAZE’s core contribution is multi-signal emotional inference: facial expression (CNN, Workshop 10), vocal emotion (MLP, Workshop 8), speech volume/RMS, response time, answer correctness, and answer text fuse with derived temporal signals into a single-inferred userstate. The implementation is face-primary: voice nudges only when the face is Neutral and the MLP clears 0.9 confidence (`fearful` excluded as the silence-attractor). GPT-5.4 generates dialogue whilst the symbolic AdaptiveEngine governs state inference, grounding output in Pepper’s affordances (Ahn et al., 2022, p. 1); a GPT-4.1 verifier handles answer-checking. Smedegaard (2019, p. 4) warns engagement reflects novelty rather than sustained interest; GAZE targets this via adaptive... TODO finish sentence (game-switching?)

2.3. Methods & Setup (35%; Alfie)

2.3.1 System Architecture

- PROOFREAD GAZE operates as a conversational loop rather than a rigid question-answer cycle. Each turn: 1) Pepper’s streamed camera frame (live `ALVideoDevice` socket, not per-turn photo) is classified for facial expression; 2) Pepper records through all four microphones with ambient-calibrated silence detection; 3) the WAV is canonicalised to mono 16 kHz by loudest-channel selection, then processed for vocal emotion, volume/RMS, and Whisper transcription; 4) the AdaptiveEngine infers user-state from face, voice, volume, response time, correctness, and answer text; 5) context plus user speech are sent to GPT-5.4 with four function-calling tools; 6) speech, gesture, and LED state fire concurrently on Pepper. Computation runs on the laptop; Pepper handles physical I/O. The dashboard hence reflects live emotional-state inference from session start.
- This function-calling architecture is neuro-symbolic: GPT-5.4 governs dialogue and decision-making, whilst the AdaptiveEngine and game logic are exposed as callable tools. This aligns with Garcez and Lamb’s (2023, p. TODO: VERIFY 12389) ‘third wave’ paradigm, wherein neural and symbolic components share a structured interface (cf. Ahn et al., 2022, p. 1).

essentially in many cases the robot will output text which controld descisiom-making (if says ‘x’ do ‘y’)

2.3.2 Input Layer: Multimodal and Behavioural Signals

1- Facial Expression (vision-based). A pre-trained CNN (Workshop 10) classifies the user’s expression into seven categories (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise) from a 48×48 greyscale face region, building upon Ekman and Friesen (1971, pp. 127-128), whose cross-cultural results show that “particular facial behaviors are universally associated with particular emotions,” finding that even preliterate (without written language) cultures with “minimal opportunity to have learned to recognize uniquely Western facial expressions” identified the same six emotions; this taxonomy remains “still the most popular perspective for FER” (Li and Deng, 2020, p. 1). Known limitations (cultural bias, resting-face misclassification) are mitigated by combining facial evidence with behavioural signals (response time, correctness, silence history, volume/RMS) rather than allowing the CNN to decide user state alone.

- PROOFREAD **2- Verbal Answer (speech-based).** Pepper records through all four microphones (the `[1,1,1,1]` mask) via `ALAudioRecorder` at 16 kHz, whilst `ALAudioDevice` polls the maximum energy across all four mics, hence catching off-axis speakers a front-only poll would miss. Recording terminates after 1.2 seconds of post-speech silence or at the adaptive hard ceiling. After `SFTP, force_mono_16k_wav()` selects the loudest channel by RMS and rewrites mono 16 kHz PCM. The canonical WAV is then transcribed via OpenAI Whisper (`whisper-1`), whose models “approach [human] accuracy and robustness” (Radford et al., 2023, Abstract). Whisper is utilised as a delivery-aware sensor: `verbose_json` exposes per-segment `no_speech_prob`, `avg_logprob`, and `compression_ratio` diagnostics; it therefore encodes *how* the user spoke alongside *what* they said cueing older ASR strips wholesale. Turn-taking is strictly sequential, wherein the mic disengages during TTS playback; barge-in (user-initiated interrupt mid-utterance) was therefore deliberately omitted, owing to NAOqi’s imperfect acoustic echo

cancellation (which would have the robot’s own voice fire false interrupts) and the resultant false-trigger risk under noisy demo conditions.

Listing 1: `nao_record`: calibrated-threshold silence detection polling the max energy across all four Pepper mics, with firmware fallback. Recording uses the `[1,1,1,1]` channel mask; `force_mono_16k_wav()` downstream picks the loudest channel by RMS so Whisper, Silero-VAD, the WS-08 MLP, and `measure_volume()` all see one canonical mono 16 kHz layout.

```

1 def nao_record(ssh, energy_threshold: int = DEFAULT_ENERGY_THRESHOLD,
2               record_max_secs: float = RECORD_MAX_SECS,
3               silence_secs: float = SILENCE_DURATION):
4     """Record audio on Pepper with dynamic silence detection.
5     - poll the max energy across all four mics to stop recording early if silence detected
6     - calibrated energy threshold to avoid false positives from ambient noise
7     - if getFrontMicEnergy is unsupported (e.g. older firmware), fall back to a safe
      ↪ fixed-duration recording to ensure the demo still works, albeit without silence
      ↪ detection
8     """
9
10    nao_run(ssh, f"""
11 from naoqi import ALProxy
12 import time
13
14 rec = ALProxy("ALAudioRecorder", "127.0.0.1", 9559)
15
16 rec.stopMicrophonesRecording()
17 rec.startMicrophonesRecording("{REMOTE_WAV}", "wav", 16000, [1, 1, 1, 1])
18
19 try:
20     audio = ALProxy("ALAudioDevice", "127.0.0.1", 9559)
21
22     speech_detected = False
23     silence_start = None
24     start = time.time()
25     threshold = {energy_threshold}
26
27     while True:
28         elapsed = time.time() - start
29
30         # hard ceiling; never exceed max duration
31         if elapsed >= {record_max_secs}:
32             break
33
34         # poll all four mics and take the MAX; a user stood to either side of Pepper registers on the
      ↪ side mics but not the front, so front-only polling misses them and the silence-detector
      ↪ stops recording mid-utterance
35         try:
36             energy = max(
37                 audio.getFrontMicEnergy(),
38                 audio.getLeftMicEnergy(),
39                 audio.getRightMicEnergy(),
40                 audio.getRearMicEnergy(),
41             )
42         except Exception:
43             energy = audio.getFrontMicEnergy() # firmware fallback
44
45         if elapsed < {RECORD_MIN_SECS}:
46             # minimum recording period
47             if energy > threshold:

```

```

48         speech_detected = True
49         time.sleep({SILENCE_POLL_SECS})
50         continue
51
52     if energy > threshold:
53         speech_detected = True
54         silence_start = None
55     else:
56         if speech_detected and silence_start is None:
57             silence_start = time.time()
58         if speech_detected and silence_start is not None:
59             if (time.time() - silence_start) >= {silence_secs}:
60                 break
61
62     time.sleep({SILENCE_POLL_SECS})
63
64 except Exception as e:
65     # firmware fallback; getFrontMicEnergy() unsupported on this Pepper
66     # fall back to a safe fixed-duration recording so the demo never breaks
67     print(" [Silence detection failed: " + str(e) + "] Falling back to fixed-duration recording")
68     time.sleep({record_max_secs})
69
70 rec.stopMicrophonesRecording()
71 """
72 sftp = ssh.open_sftp()
73 sftp.get(REMOTE_WAV, LOCAL_WAV)
74 sftp.close()

```

- PROOFREAD Whisper hallucinates on near-silent audio, emitting training-set tics (CJK gibberish, prompt-primed repetition); the latter NLG *degeneration* wherein models get “stuck in repetitive loops” (Ji et al., 2023, p. 3). Transcription is gated by five layers: 1) a near-empty-file RMS floor (NAO_MIN_RMS_TO_TRANSCRIBE = 30, emergency only; the older 120 wrongly rejected short answers like “yes” or “Tom”); 2) Silero-VAD, applied to both LOCAL and NAO modes since `force_mono_16k_wav()` canonicalises beforehand; 3) Vosk grammar-restricted wake-word gate (“Pepper”/“Gaze”), bypassable for the first-turn name prompt; 4) Whisper’s own `no_speech_prob` / `avg_logprob` / `compression_ratio` signals; 5) an aggressively-normalised hallucination blacklist.

Listing 2: `transcribe`: five-layer defence chain against Whisper hallucination on near-silent audio (`gaze.py`, lines 1357–1427).

```

1 def transcribe(bypass_wake_word: bool = False,
2               whisper_prompt: str = "User answers a quiz or chats with a companion robot.") -> str:
3     """
4     Transcribe the local WAV with Whisper via a five-layer defence:
5     1- existing RMS + speech_detected pre-gate (in conversation_loop)
6     2- Silero VAD hard gate (this function; both LOCAL and NAO modes)
7     3- Vosk wake-word gate (requires "Pepper"/"Gaze"; bypassable)
8     4- Whisper's own no_speech_prob + avg_logprob (from verbose_json)
9     5- aggressively-normalised hallucination blacklist
10
11     Returns "" on failure, on any gate rejecting, or when the
12     transcription normalises to a known-hallucination phrase.
13
14     bypass_wake_word=True disables Layer 3 for turns where no wake-word
15     is expected (e.g. the first-turn name prompt).
16     """
17     if LOCAL_MODE and not _local_speech_detected:
18         return ""
19

```

```

20 # Layer 2: Silero VAD hard gate; applied to both LOCAL and NAD modes because
    ↪ force_mono_16k_wav() canonicalises the WAV before transcribe() runs
21 if not has_real_speech(LOCAL_WAV):
22     print(" Silero VAD found no speech; skipping Whisper.")
23     return ""
24
25 # Layer 3: Vosk wake-word gate. User must say "Pepper"/"Gaze"; bypassable for the first-turn
    ↪ name prompt where no wake-word is expected.
26 if not bypass_wake_word and not has_wake_word(LOCAL_WAV):
27     print(" No wake-word detected; skipping Whisper.")
28     return ""
29
30 try:
31     with open(LOCAL_WAV, "rb") as fh:
32         resp = client.audio.transcriptions.create(
33             model="whisper-1",
34             file=fh,
35             response_format="verbose_json",
36             temperature=0.0,
37             prompt=whisper_prompt,
38             timeout=API_TIMEOUT,
39         )
40     text = (getattr(resp, "text", "") or "").strip()
41
42 # Layer 4: Whisper self-signals (no_speech_prob + avg_logprob + compression_ratio per
    ↪ segment). resp is a Pydantic TranscriptionVerbose thus attribute access (not dict
    ↪ indexing).
43 segments = getattr(resp, "segments", None) or []
44 if segments:
45     no_speech_vals = [s.no_speech_prob for s in segments
46                       if getattr(s, "no_speech_prob", None) is not None]
47     logprob_vals = [s.avg_logprob for s in segments
48                    if getattr(s, "avg_logprob", None) is not None]
49     compression_vals = [s.compression_ratio for s in segments
50                        if getattr(s, "compression_ratio", None) is not None]
51     if no_speech_vals and max(no_speech_vals) > 0.6:
52         print(f" Whisper flagged silence (max no_speech_prob="
53               f"{max(no_speech_vals):.2f}); dropping {text!r}")
54         return ""
55     if logprob_vals and min(logprob_vals) < -1.0:
56         print(f" Whisper low-confidence (min avg_logprob="
57               f"{min(logprob_vals):.2f}); dropping {text!r}")
58         return ""
59     # compression_ratio > 2.4 catches repetition loops (e.g. "Q1. Q1. Q1..."); repeating
    ↪ tokens compress well. 2.4 matches local Whisper's default.
60     if compression_vals and max(compression_vals) > 2.4:
61         print(f" Whisper repetition loop (max compression_ratio="
62               f"{max(compression_vals):.2f}); dropping {text!r}")
63         return ""
64
65 # Layer 5: aggressively-normalised hallucination blacklist
66 if is_known_hallucination(text):
67     print(f" Filtered Whisper hallucination: {text!r}")
68     return ""
69
70 # Strip leading "Pepper"/"Gaze" so handlers receive just the answer; \b blocks
    ↪ "Pepperoni"/"Gazebo" false positives..
71 stripped = re.sub(r'(?i)^\s*(pepper|gaze)\b[,\.\s]*', '', text).strip()
72 return stripped

```

```

73     except Exception as e:
74         print(f" Whisper transcribe failed ({e}); returning empty")
75         return ""

```

3- Vocal Emotion (audio-based). The same WAV is passed through a pre-trained MLP (Workshop 8) *before* transcription, classifying vocal state into four emotions (calm, happy, fearful, disgust) via MFCC, chroma, and mel-spectrogram features; El Ayadi, Kamel and Karray (2011, p. 577) identify MFCCs as “the most promising features” for speech-emotion recognition. This provides a second, independent modality; the two may disagree, wherein decision logic arbitrates. RAVDESS is an acted-speech corpus, and indeed Williams and Stevens (cited in El Ayadi et al., 2011, p. 573) found “acted emotions tend to be more exaggerated than real ones”. The WS-08 feature pipeline mean-pools MFCC, chroma, and mel across the utterance into a single global vector (*i.e. one fixed-size summary per clip*), wherein “temporal information present in speech signals is completely lost” (ibid., p. 574); fear’s defining tremor and irregular-pitch structure consequently collapses, indistinguishable from low-energy real-room speech. Voice is therefore engineered as a tie-breaker (the gate is detailed in Section 2.3.3), an architectural response to a documented limitation rather than a debugging afterthought.

Listing 3: `SpeechEmotionModel.extract_features`: MFCC + chroma + mel-spectrogram feature vector matching the WS-08 training pipeline, with peak-normalisation for quieter brain-injured users (gaze.py, lines 192–220).

```

1
2 # PASTE SpeechEmotionModel.extract_features FROM gaze.py HERE (lines ~192--220, including the
   ↪ @staticmethod decorator)

```

4- Response Time (engagement-based). A Python timer measures elapsed time from question delivery to recording completion; the Whisper call occurs *after* the timer halts, isolating deliberation time from API latency.

5- Answer Correctness (task-based). GPT-4.1’s `check_game_answer` evaluates the transcribed answer at a deterministic temperature; the resultant binary correctness feeds the AdaptiveEngine and the derived rolling-correctness signal.

2.3.3 Process Layer: Multi-Signal State Inference

- PROOFREAD The AdaptiveEngine’s `infer_state()` combines multimodal and behavioural signals to classify the user into one of five states: *Thriving*, *Comfortable*, *Struggling*, *Frustrated*, or *Disengaged*. Five main input channels are supplemented by three derived temporal features: rolling correctness over the last five rounds, consecutive-silence count, and consecutive-wrong streak length. Crucially, classification is face-primary: facial expression, correctness, response time, silence, wrong-streaks, and volume carry the decision; vocal emotion fires only as a high-confidence tie-breaker when face is Neutral AND the MLP clears 0.9 AND the label is not **fearful** (the silence-attractor; RAVDESS’s actor-exaggerated fear collides with low-energy real-room speech in the mean-pooled feature-space, hence the gate). Voice is thus retained as evidence but never allowed to override stronger behavioural readings, addressing the brittleness Desai et al. (2013, p. 256) observe in single-signal systems. Thresholds (correctness floor 0.4, ceiling 0.8, response-time baseline 15s, consecutive-wrong trigger 3, silence threshold 2) were derived from pilot testing.

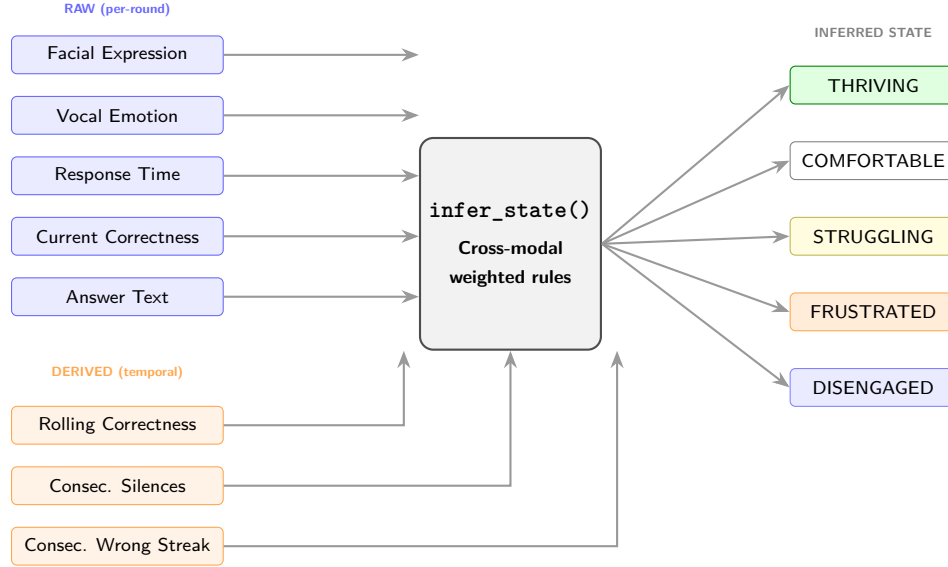


Figure 4: Multi-signal inference pipeline. Five raw inputs captured each round and three derived temporal signals computed from session history feed into `infer_state()`, which applies cross-modal rules (e.g. camera reads *Angry* but user answers fast and correctly $\rightarrow$ *Comfortable*) to classify the user into one of five states. Voice is treated as advisory rather than dominant: it nudges the state only when the face is Neutral AND the MLP’s confidence clears 0.9 AND the label is not *fearful* (the silence-attractor), hence a noisy vocal label can never override stronger visual or behavioural evidence; therein lies the multi-signal novelty.

The complete classification logic, face-primary with voice tie-breakers, is shown in Listing~4 (extracted from `gaze.py infer_state()`):

Listing 4: Multi-signal inference rules (verbatim excerpt from `gaze.py infer_state()`; method-body indentation stripped). The newest implementation is face-primary: every rule fires off facial expression + behavioural signals first, voice is consulted only as a high-confidence tie-breaker at the end.

```

1
2 # 1- FACE-PRIMARY RULES: these fire before voice is ever consulted
3
4 # thriving: performing well + fast responses
5
6 if (correctness >= CORRECTNESS_CEILING and response_time < RESPONSE_TIME_BASELINE * 0.5):
7     return InferredState.THRIVING
8 if expression == "Angry" and correct and response_time < RESPONSE_TIME_BASELINE * 0.6:
9     return InferredState.COMFORTABLE
10
11 # disengaged: silence threshold, or face + slow response + poor performance
12
13 if self.consecutive_silences >= SILENCE_THRESHOLD:
14     return InferredState.DISENGAGED
15 if (expression == "Neutral"
16     and response_time > RESPONSE_TIME_BASELINE
17     and correctness < 0.5):
18     return InferredState.DISENGAGED
19 if (low_arousal and expression == "Neutral"
20     and response_time > RESPONSE_TIME_BASELINE * 0.8):
21     return InferredState.DISENGAGED
22
23 # frustrated: negative face + poor performance, or 3+ wrong + negative face, or high arousal +
    ↪ negative face + poor

```

```

24
25 if expression in ("Angry", "Disgust") and correctness < CORRECTNESS_FLOOR:
26     return InferredState.FRUSTRATED
27 if self.consecutive_wrong >= 3 and expression in ("Angry", "Sad", "Fear"):
28     return InferredState.FRUSTRATED
29 if (high_arousal and expression in ("Angry", "Disgust", "Fear")
30     and correctness < CORRECTNESS_FLOOR):
31     return InferredState.FRUSTRATED
32
33 # struggling: sadness + slow; poor correctness; fear + wrong
34
35 if expression == "Sad" and response_time > RESPONSE_TIME_BASELINE * 0.7:
36     return InferredState.STRUGGLING
37 if correctness < CORRECTNESS_FLOOR:
38     return InferredState.STRUGGLING
39 if expression == "Fear" and not correct:
40     return InferredState.STRUGGLING
41
42 # 2- VOICE TIE-BREAKERS: only when face is Neutral AND voice is trustworthy. Voice can nudge but
43     ↪ never override a face-driven verdict.
44
45 if expression == "Neutral" and trust_voice:
46     if vocal_emotion == "happy" and correct and correctness >= CORRECTNESS_CEILING:
47         return InferredState.THRIVING
48     if vocal_emotion == "calm" and correctness >= 0.5:
49         return InferredState.COMFORTABLE
50
51 return InferredState.COMFORTABLE

```

The `decide()` function then maps the inferred state to concrete adaptive actions: difficulty adjustment (easy, medium, hard), game switching (numbers ↔ letters when the user is frustrated or disengaged), hint provision, encouragement, and tone selection. This parallels Nikolaidis, Hsu and Srinivasa’s (2017, p. 625) mutual-adaptation paradigm. Table~1 summarises the mapping.

Inferred State	Difficulty	Game Switch	Hint	Tone	LED Colour
Thriving	↑ ramp up	No	No	Energetic	Green
Comfortable	↑ if >70%	No	No	Neutral	Cyan
Struggling	↓ ease off	No	Yes	Encouraging	Yellow
Frustrated	↓↓ Easy	After 4 wrong	No	Calm	Red
Disengaged	—	After 3 checkouts	No	Energetic	Magenta

Table 1: State-to-action mapping. The adaptive engine translates each inferred state into a specific combination of difficulty adjustment, game-type switch, hint provision, tone selection, and LED colour. Arrows indicate direction of difficulty change relative to the current level, wherein the mapping applies solely to `decide()`’s engine output (the spoken-checkout path, e.g. repeated *skip*); the main-loop tiered intervention on pure-silence turns, tier-1 check-in at 3 silences and tier-2 switch at 4, bypasses `decide()` entirely.

Disengagement is flagged by three OR’d rules wherein any one trips it: 1) two consecutive silences (mic-empty or “skip”/“pass” checkout phrases), 2) neutral-face + response > 15s + rolling-correctness < 50% (the cognitive-checkout pattern), 3) quiet-voice + neutral-face + slow-ish response (early-warning drift, catches it before accuracy tanks). GAZE then tiers its intervention, gentle check-in at three silences, game-switch at four, wherein each tier fires at most once per silent spell.

2.3.4 Generate Layer: Dynamic Prompt Construction

- PROOFREAD
- Rather than constructing a fixed game prompt each round, GAZE utilises OpenAI function calling to let GPT-5.4 decide actions. Every turn, `build_signal_context()` packages live signals (face, voice, volume, response time, rolling accuracy, recent-face/recent-vocal windows) into a context block prepended to the user’s transcribed speech. This is sent to GPT-5.4 with four callable tools: `generate_game_question`, `check_game_answer`, `evaluate_last_adaptation`, and `request_more_time`. The LLM decides which to invoke; during natural conversation it calls none, whilst during gameplay it sequences `check_game_answer` and `generate_game_question`. Game-question generation runs as a separate JSON-only call with a variety seed plus a rolling 30-question do-not-repeat memory, hence preventing mode-collapsed targets. Answer verification is delegated to a deterministic GPT-4.1 verifier (temperature 0.0), reducing the hallucination risk Ji et al. (2023, p. 3) identify. A 10-second timeout wraps every API call in case the network stalls so Pepper falls back gracefully rather than freezing mid-turn.

2.3.5 Output Layer: Aligned Multimodal Response

Speech, gestures, and LED state fire concurrently via threading. Context-aligned gestures (e.g. animated speech for celebratory moments) execute alongside dialogue, whilst LED colours reflect the inferred state, providing a secondary non-verbal feedback channel.

2.3.6 Adaptation Self-Evaluation and Session Persistence

- PROOFREAD GAZE exposes `evaluate_adaptation()` to the LLM as the `evaluate_last_adaptation` tool, hence the LLM can compare consecutive rounds to judge whether a previous adaptation helped (e.g. did a difficulty decrease after frustration produce a correct answer?). Session progress is saved to `game_save.json` after every game round, with a belt-and-braces auto-save every two turns regardless, in case the session is interrupted before the next scheduled save.

2.4. Outcome & System Analysis (30%; Salman)

2.5. Conclusion (10%; Alfie)

- PROOFREAD GAZE implements multi-signal emotional inference across two independent modalities (*facial expression via WS-10 CNN and vocal emotion via WS-08 MLP*) alongside speech volume/RMS, response time, answer correctness, answer text, and derived temporal signals, hence yielding a more robust user-state estimate than any single channel. The implementation hardens further by recording all four Pepper microphones, canonicalising audio to mono 16 kHz by loudest-channel selection, applying Silero-VAD to both modes, and streaming Pepper’s camera via persistent `ALVideoDevice` rather than per-turn SFTP. The hybrid architecture (GPT-5.4 dialogue paired with symbolic rule-based inference and a deterministic GPT-4.1 verifier) and adaptive game-switching directly target the novelty-decay problem Smedegaard (2019, p. 4) identifies.
- PROOFREAD The conversational architecture, wherein the LLM decides actions via function calling, positions GAZE as a social companion rather than a rigid game host. <!-- WRAPPED OUT: factually
→ inaccurate (no personality-mode selection in current gaze.py)The selectable personality
→ modes extend Tapus, Tapus and Mataric's (2008, p. TODO: VERIFY PAGE)personality matching
→ from static trait-matching to dynamic, signal-driven adaptation. -->

The multi-signal approach could transfer to stroke rehabilitation re-engagement (Mataric et al., 2007), educational tutoring, or neurodivergent support. Future work could replace hand-coded thresholds with learned parameters from longitudinal data, and fine-tune both models on in-session Pepper captures.

2.6 Task-4 References (5%)

- verify all references

Alfie's

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2.7. Appendix

2.7.1. Code

2.7.2. Video Demo

- YouTube link: test

Appendices

Appendix A: AI Declaration

Solo Work	S1 - Generative AI tools have not been used for this assessment.
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

Figure 5: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

AI Tool Used	Purpose of Use	Extent of Use
ChatGPT	Brainstorming project ideas and structuring the report (A2)	Initial brainstorming and final outline stages
ChatGPT	Reviewing structural alignment against grading criteria and mapping word-count budgets (A2)	After and midway through section-drafting
ChatGPT	Finding relevant pages to read in the paper (A4)	Few times if the paper is too long
ChatGPT	General conversations via web-search AI about prevalent papers to read about how the topics relates to others' studies (A4)	Few times at the end

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provided above are an accurate description of how AI was used for this assessment.